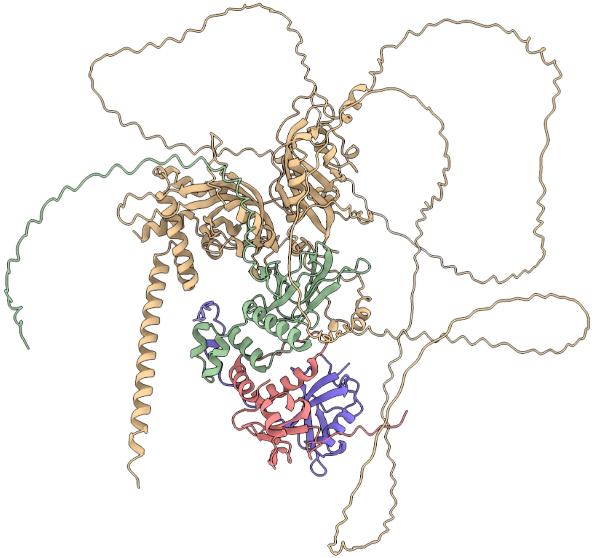
AF-Pipeline

Package to assist in AlphaFold2[1] and AlphaFold3[2] related tasks

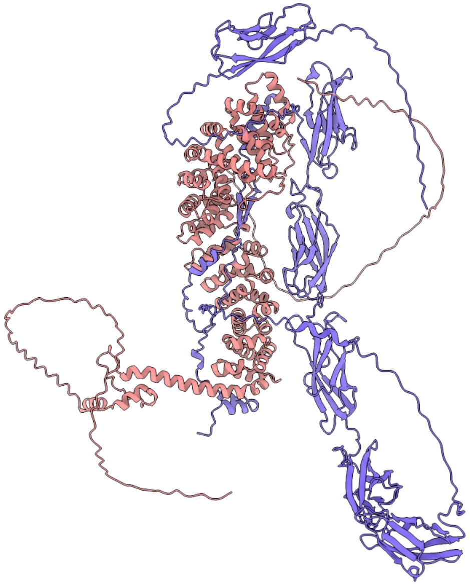
Why do we need AF-Pipeline?

HIF-α bound to pVHL/elongin-C/elongin-B complex



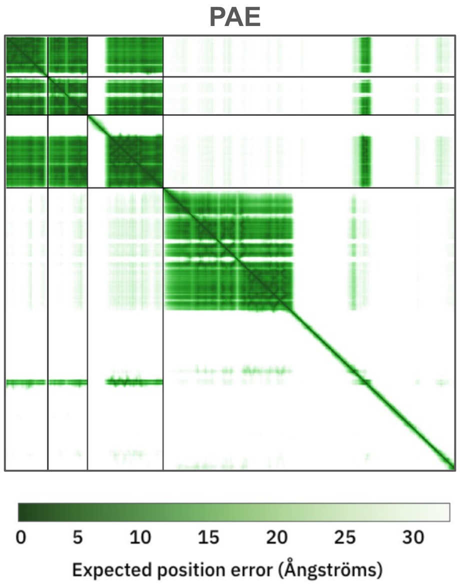
ipTM = 0.67
pTM = 0.4

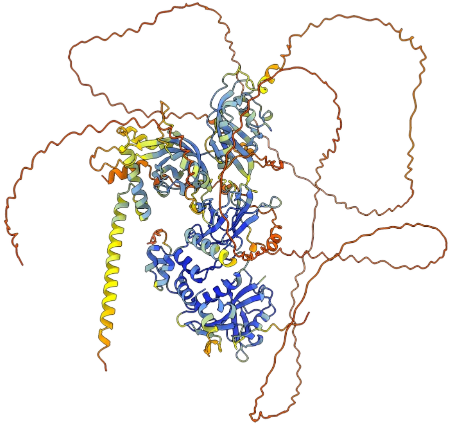
Cadherin-catenin complex

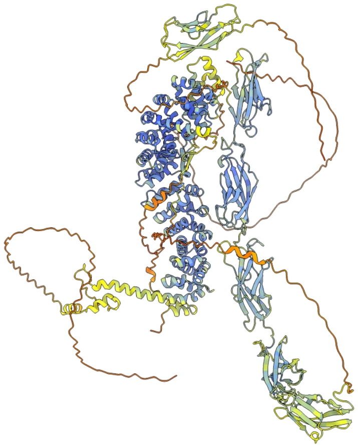


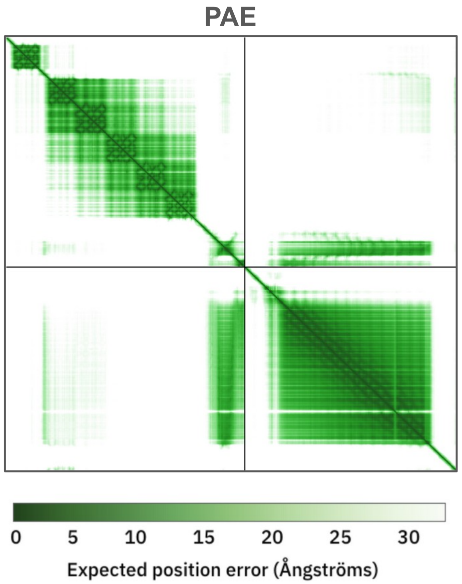
ipTM = 0.61
pTM = 0.46

Why do we need AF-Pipeline?









ipTM = 0.67
pTM = 0.4

pLDDT score 0 | 50 | 70 | 90 | 100

100	to 90	high accuracy expected
90	to 70	backbone expected to be modeled well
70	to 50	low confidence, caution
50	to 0	should not be interpreted, may be disordered

ipTM = 0.61
pTM = 0.46

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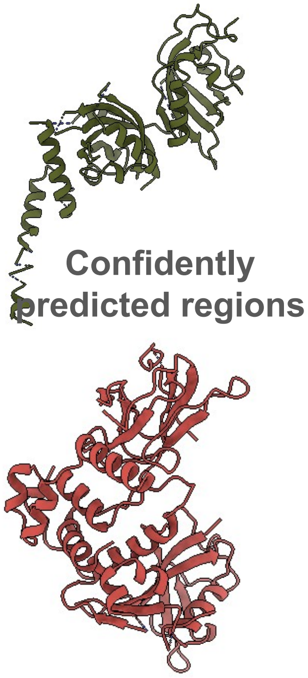
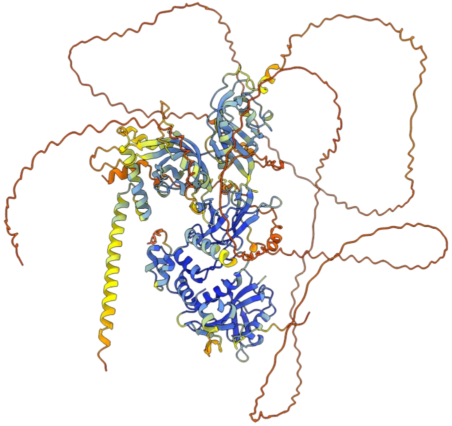
HIF-α bound to pVHL/elongin-C/elongin-B complex

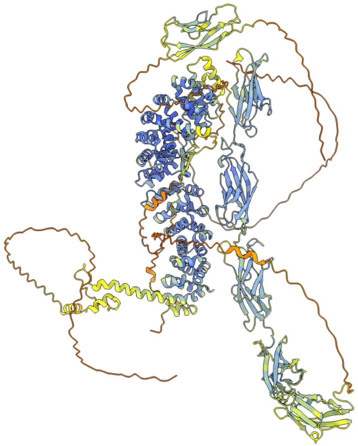


Confidently predicted regions

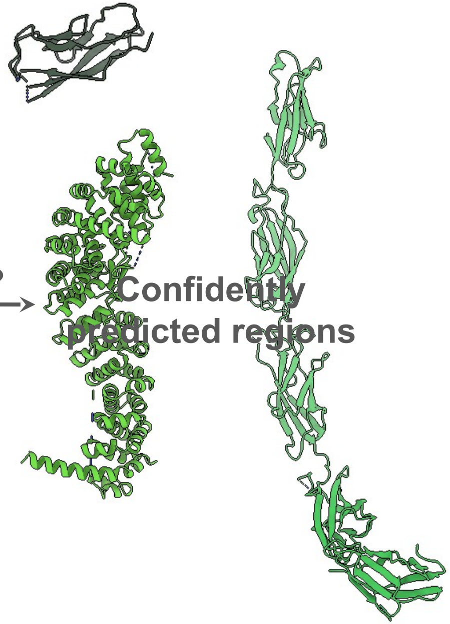
?



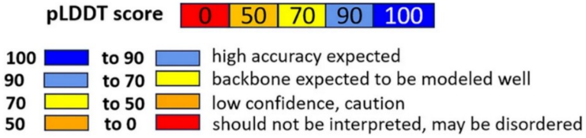
Cadherin-catenin complex



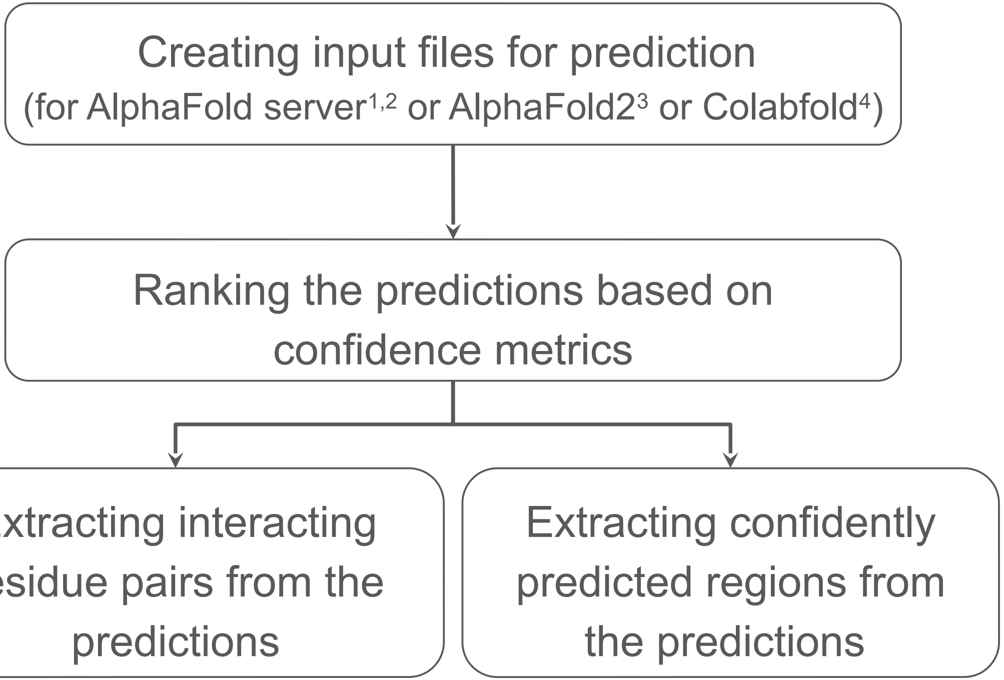
?



Confidently predicted regions

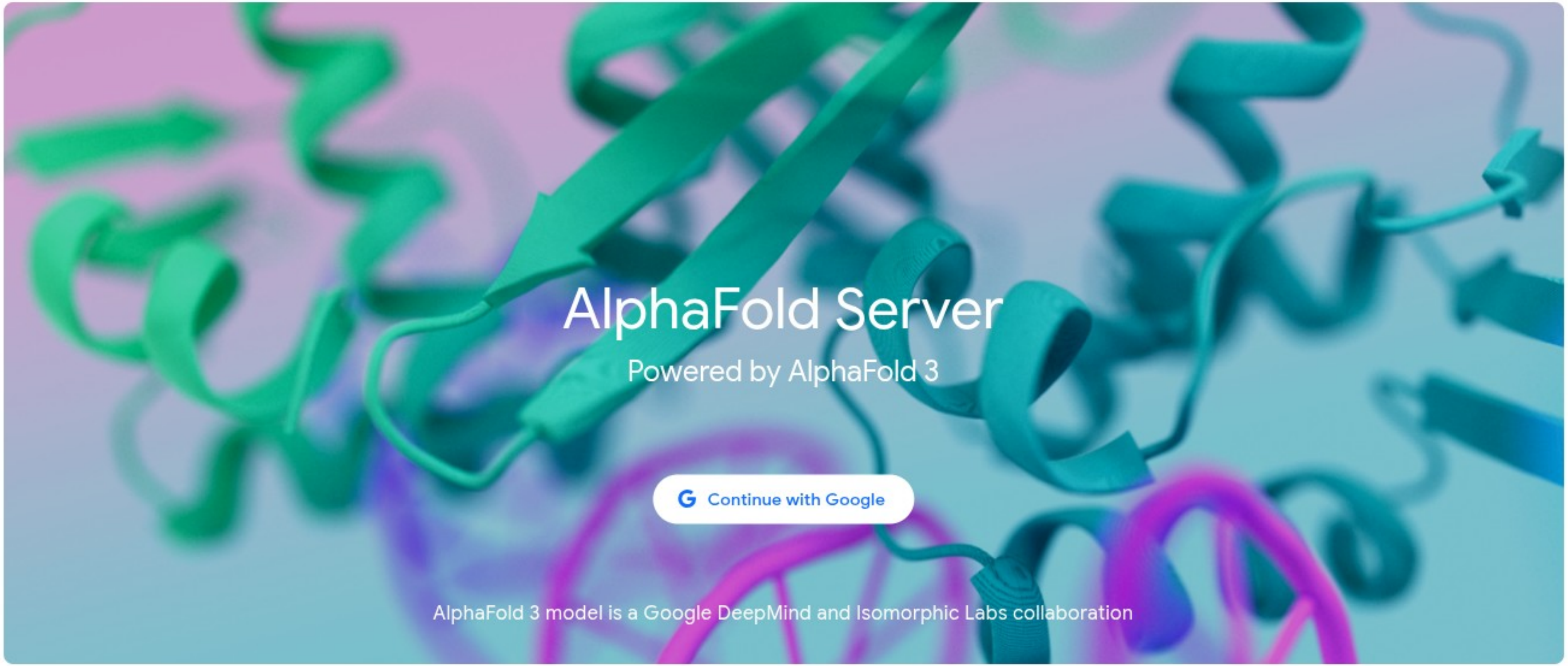


AF-Pipeline Workflow

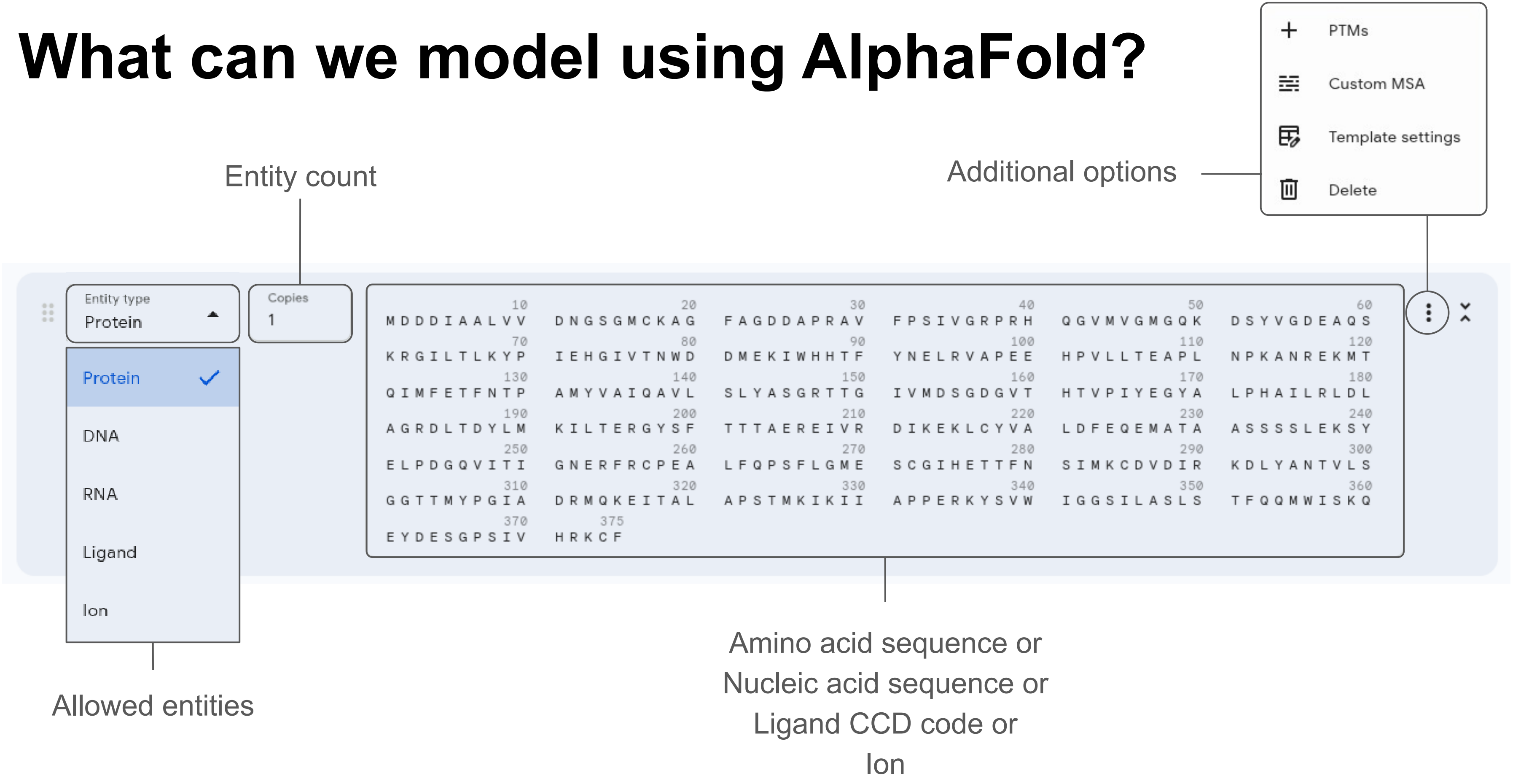


1. AlphaFold Server. alphafoldserver.com. Available at: https://alphafoldserver.com/.
2. Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493–500 (2024). (https://alphafoldserver.com/)
3. Jumper, J. et al. Highly Accurate Protein Structure Prediction with Alphafold. Nature 596, 583–589 (2021).
4. Mirdita, M. et al. ColabFold: making protein folding accessible to all. Nature Methods 19, 679–682 (2022). (https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb)

AlphaFold server



What can we model using AlphaFold?



```json
[
    {
        "name": "actin_monomer",
        "modelSeeds": [
            1
        ],
        "sequences": [
            {
                "proteinChain": {
                    "sequence":
                    "MDDDIAALVVDNGSGMCKAGFAGDDAPRAVFPSIVGRPRHQGVMVGMGQKDSYVGDEAQ
                    SKRGILTLKYPIEHGIVTNWDDMEKIWHHTFYNELRVAPEEHPVLLTEAPLNPKANREKM
                    TQIMFETFNTPAMYVAIQAVLSLYASGRTTGIVMDSGDGVTHTVPIYEGYALPHAILRLD
                    LAGRDLTDYLMKILTERGYSFTTTAEREIVRDIKEKLCYVALDFEQEMATAASSSSLEKS
                    YELPDGQVITIGNERFRCPEALFQPSFLGMESCGIHETTFNSIMKCDVDIRKDLYANTVL
                    SGGTTMYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIGGSILASLSTFQQMWISK
                    QEYDESGPSIVHRKCF",
                    "glycans": [],
                    "modifications": [],
                    "count": 1,
                    "useStructureTemplate": true,
                    "maxTemplateDate": "2021-09-30"
                }
            }
        ]
    }
]
```

AF-Pipeline config

```yaml
protein_uniprot_map:
  Act1: "P60709"

af_input_jobs:
  - modelSeeds: [1]
    entities:
    - name: "Act1"
      type: "proteinChain"
```

https://github.com/isblab/af_pipeline/blob/main/examples/input/config.yaml

```json
[
    {
        "name": "actin_monomer",
        "modelSeeds": [
            1
        ],
        "sequences": [
            {
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                    SKRGILTLKYPIEHGIVTNWDDMEKIWHHTFYNELRVAPEEHPVLLTEAPLNPKANREKM
                    TQIMFETFNTPAMYVAIQAVLSLYASGRTTGIVMDSGDGVTHTVPIYEGYALPHAILRLD
                    LAGRDLTDYLMKILTERGYSFTTTAEREIVRDIKEKLCYVALDFEQEMATAASSSSLEKS
                    YELPDGQVITIGNERFRCPEALFQPSFLGMESCGIHETTFNSIMKCDVDIRKDLYANTVL
                    SGGTTMYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIGGSILASLSTFQQMWISK
                    QEYDESGPSIVHRKCF",
                    "glycans": [],
                    "modifications": [],
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                }
            }
        ]
    }
]
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```

Open in Colab

https://github.com/isblab/af_pipeline

AlphaFold server prediction output

- fold_nap1_2_1t391_17_full_data_0.json
- fold_nap1_2_1t391_17_full_data_1.json
- fold_nap1_2_1t391_17_full_data_2.json
- fold_nap1_2_1t391_17_full_data_3.json
- fold_nap1_2_1t391_17_full_data_4.json

pLDDT
PAE
Contact probabilities

- fold_nap1_2_1t391_17_model_0.cif
- fold_nap1_2_1t391_17_model_1.cif
- fold_nap1_2_1t391_17_model_2.cif
- fold_nap1_2_1t391_17_model_3.cif
- fold_nap1_2_1t391_17_model_4.cif

Predicted structures

- fold_nap1_2_1t391_17_summary_confidences_0.json
- fold_nap1_2_1t391_17_summary_confidences_1.json
- fold_nap1_2_1t391_17_summary_confidences_2.json
- fold_nap1_2_1t391_17_summary_confidences_3.json
- fold_nap1_2_1t391_17_summary_confidences_4.json

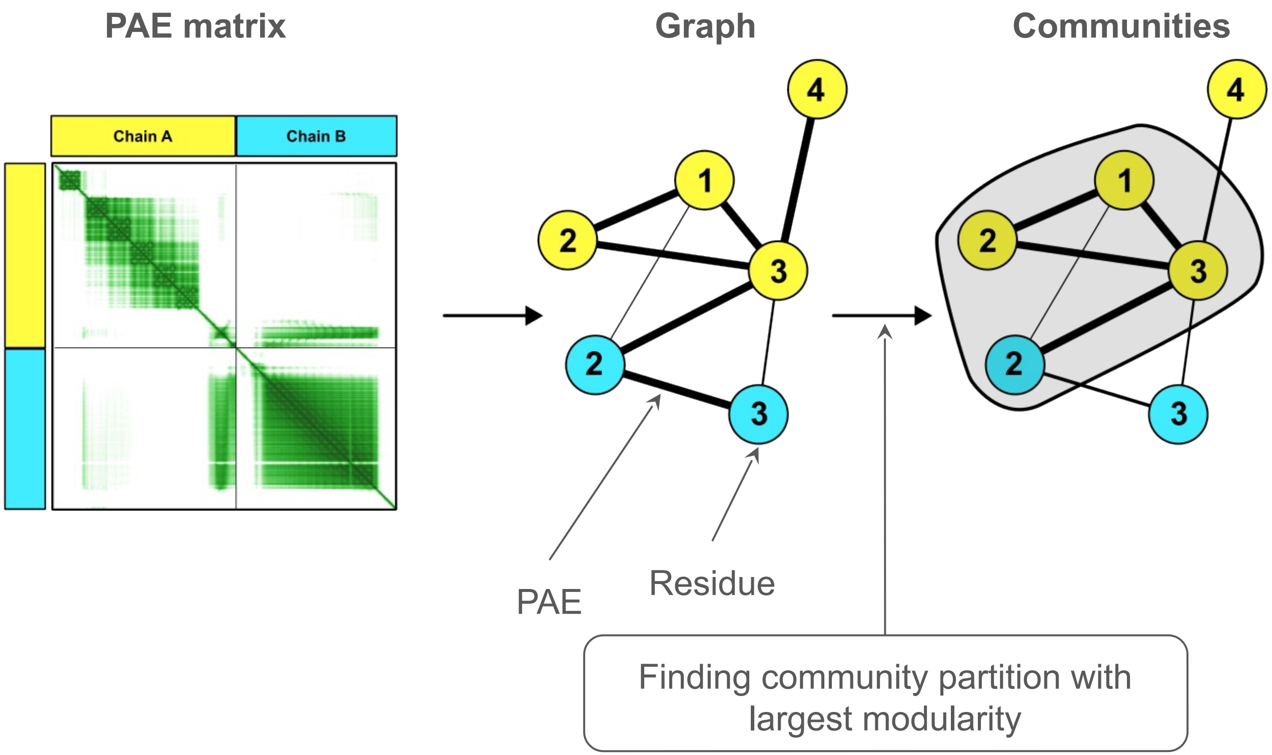
ipTM
pTM
Ranking score

- fold_nap1_2_1t391_17_job_request.json
- msas
- templates

Identifying confident residue-pair interactions

- Distance between the representative atoms ≤ 8 Å
- **Representative atoms:**
  - Proteins: Cβ or Cα (in case Cβ is absent)
  - DNA or RNA: C4 in case of purines and C2 in case of pyrimidine
  - Ion
  - Ligand ??

Extracting confidently predicted regions



https://github.com/tristanic/pae_to_domains

THANK YOU

Presenters:
Kartik Majila
Omkar Golatkar